

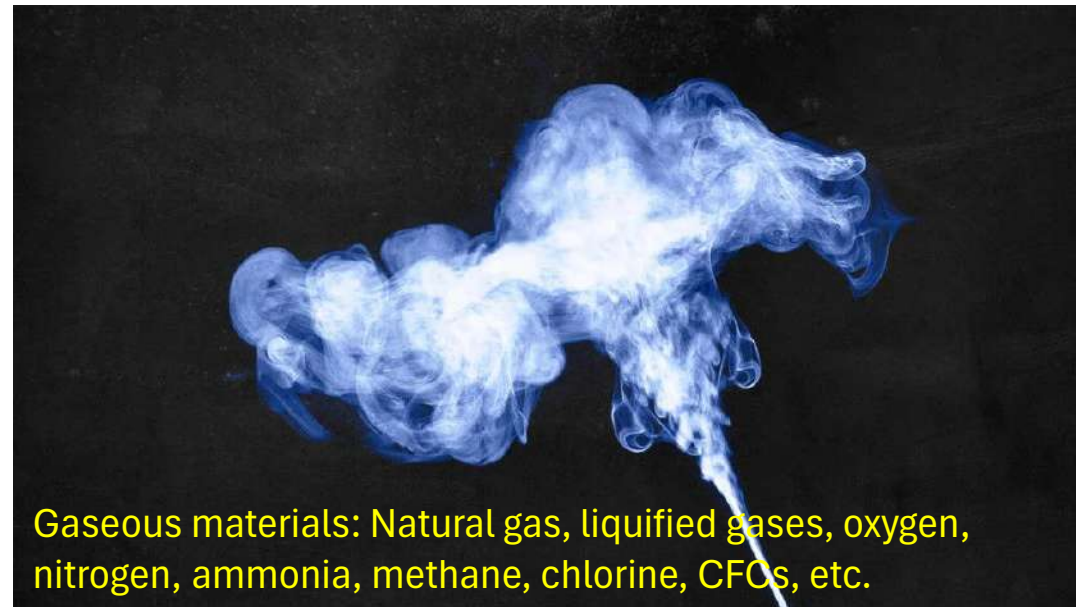
Lecture on

Robotic Manufacturing



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Classification of materials



Solids form the majority of the materials which are handled in industrial situations. Solids are classified into two groups: Unit load and Bulk load.

- **Unit loads** are formed solids of various shapes, sizes, and weights. Some of these are counted by numbers such as machine parts, molding boxes, cranes, etc. Tared goods like containers, bags, packaged items, and also the materials that are handled en-masses like forest logs, iron billets, etc. are examples of unit load.
- **Bulk materials** are those which are powdery, granular, or lumpy in nature and are stored in heaps. Examples include ores, coals, gravel, sand, clay, cement, food grains, sugar, salt, etc.





Bulk material handling



Unit load handling: Industrial trucks

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Industrial trucks

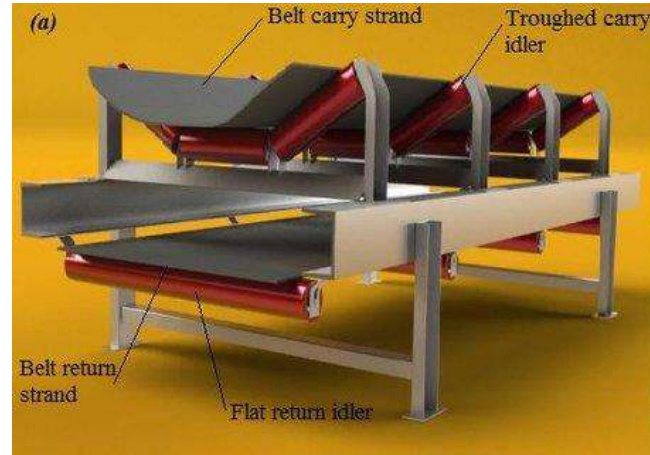


Conveyor system

Belt conveyor



Flat belt conveyor



Troughed belt conveyor



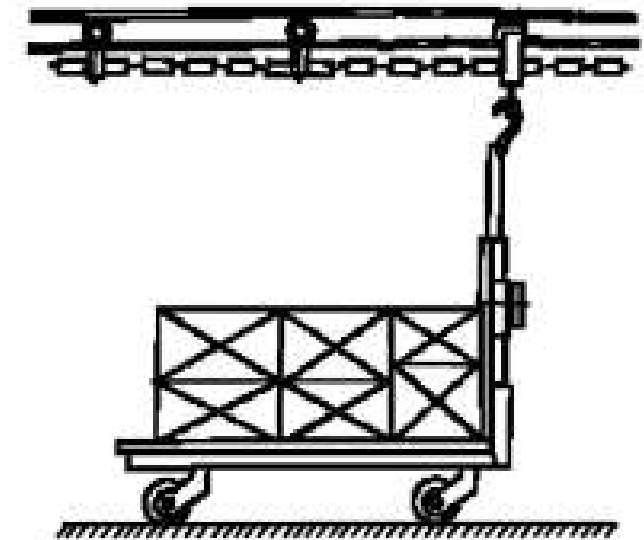
Closed belt conveyor



Roller conveyor

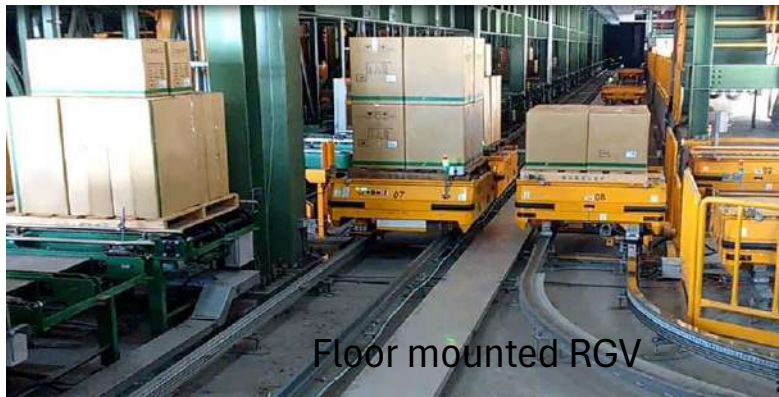


Chain conveyor

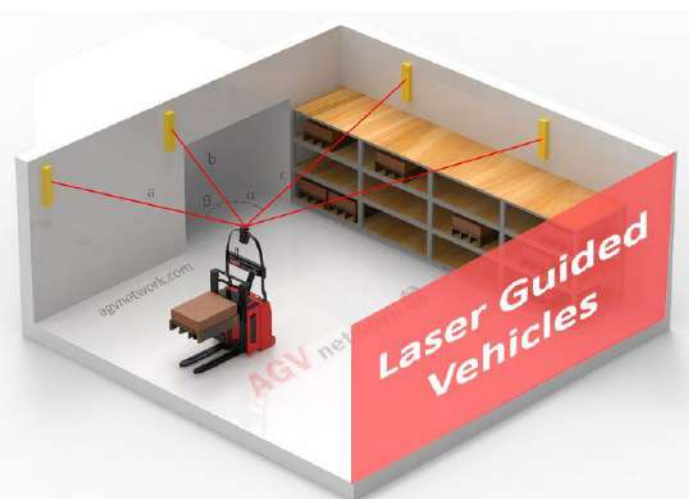


Tow-line conveyor

Rail guided vehicles (RGV)



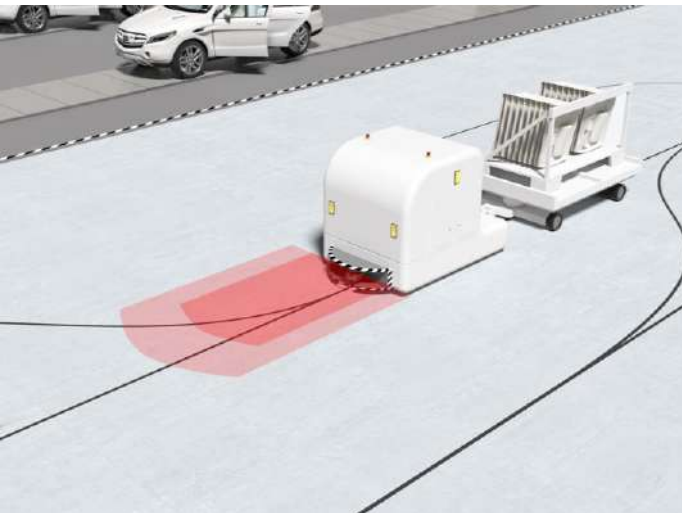
Automated guided vehicles (AGV)



Laser-Guided AGV:
These AGVs use laser scanners to detect reflectors placed along the floor or in the environment. The laser sensors map the environment and guide the vehicle's movement.



Magnetic tape guided AGV:
These AGVs follow a magnetic strip or inductive wire embedded in the floor. The vehicle detects the magnetic field and follows the path accordingly.



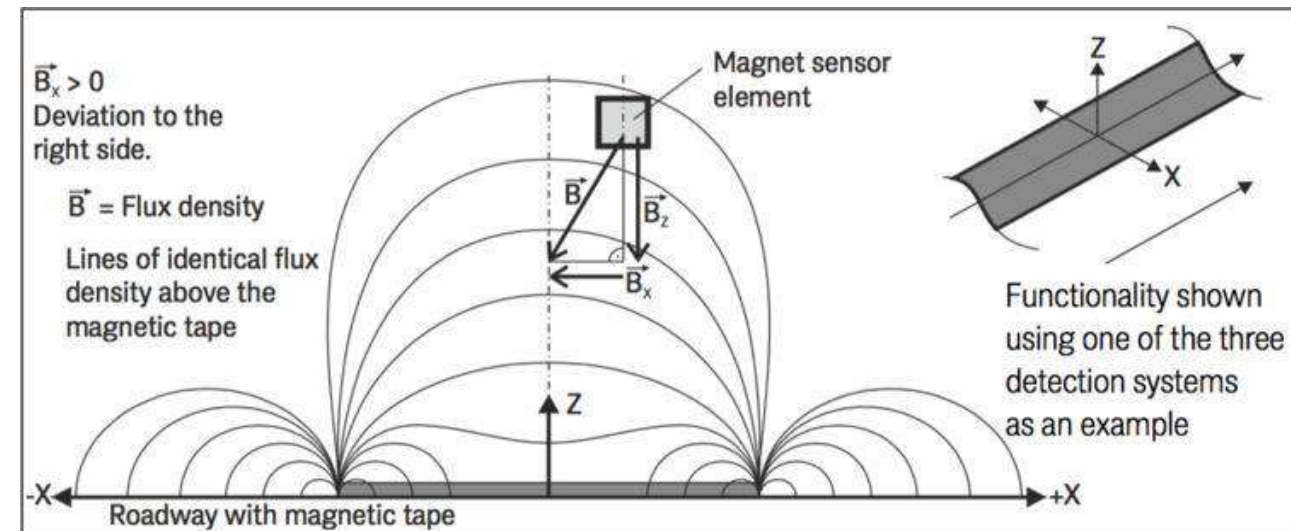
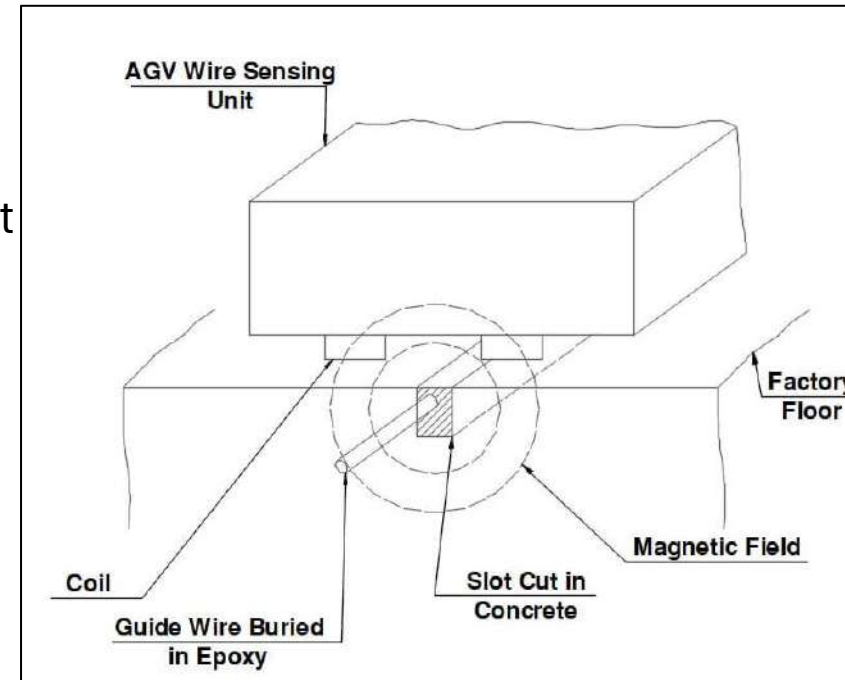
Camera-Guided AGV:
Uses cameras or optical sensors to detect natural or artificial markings (e.g., QR codes, reflective tape, or floor lines) on the floor.



Vision-Guided AGV:
Uses computer vision and image processing techniques to detect obstacles and navigate. These systems use cameras or LIDAR to build a real-time map of the environment.

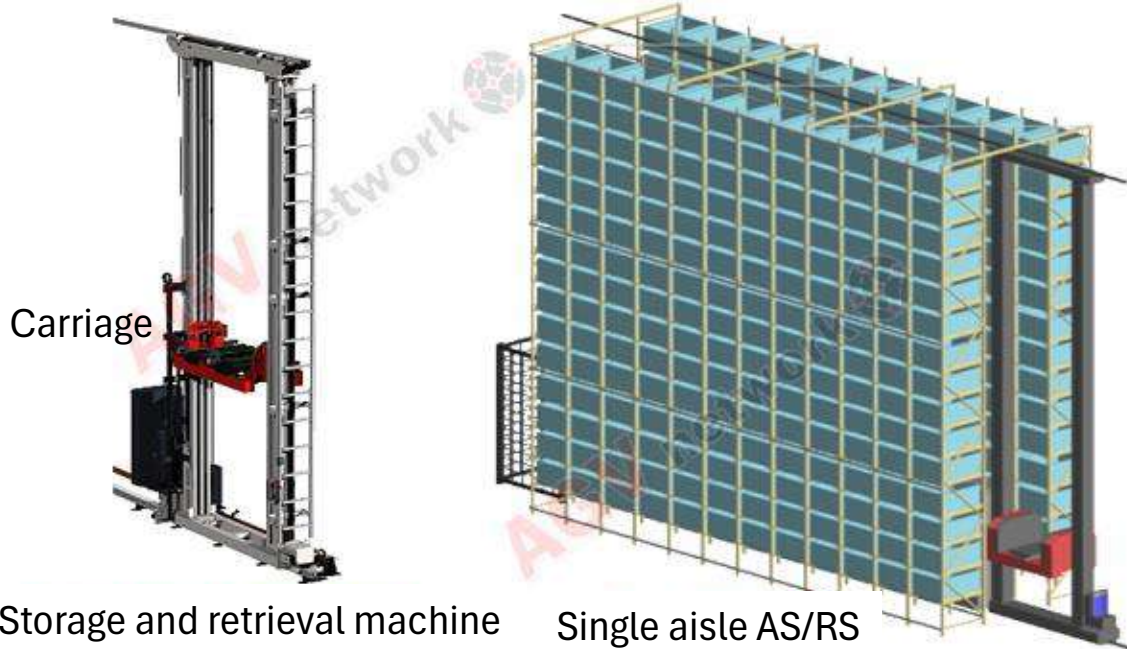
Working principle of inductive wire-based AGV

- A copper or aluminum wire is embedded in the floor or mounted along the pathway, typically under the floor surface.
- This wire generates a magnetic field when an electric current flows through it. The path of the wire defines the route that the AGV must follow.
- The electric current passing through the wire creates an electromagnetic field.
- The AGV is equipped with inductive sensors (usually Hall effect sensors or magnetic encoders) that are designed to detect the magnetic field produced by the wire. These sensors are usually located underneath or on the sides of the vehicle.
- The sensors detect changes in the strength of the magnetic field as the AGV moves along the path.
- If the AGV drifts to the left or right, the magnetic field detected by the sensors on one side becomes stronger than the other. The controller adjusts the steering mechanism (typically by changing the speed of the wheels on one side) to correct the vehicle's position and keep it aligned with the path.



Automated Storage and Retrieval System (AS/RS)

- It is a storage system that performs storage and retrieval operations with speed and accuracy.
- An aisle (passage) is usually kept between the rack structures.
- S/R machine handles the unit loads for storing into the rack and retrieving from the rack.
- Three movements involve => (i) horizontal movement of the S/R machine, (ii) Vertical movement of the carriage, (iii) shuttle transfer between the carriage and the storage compartment.
- Parallel rail at top of the storage structure is used to maintain alignment of the carriage with the rack structure.



Single aisle AS/RS

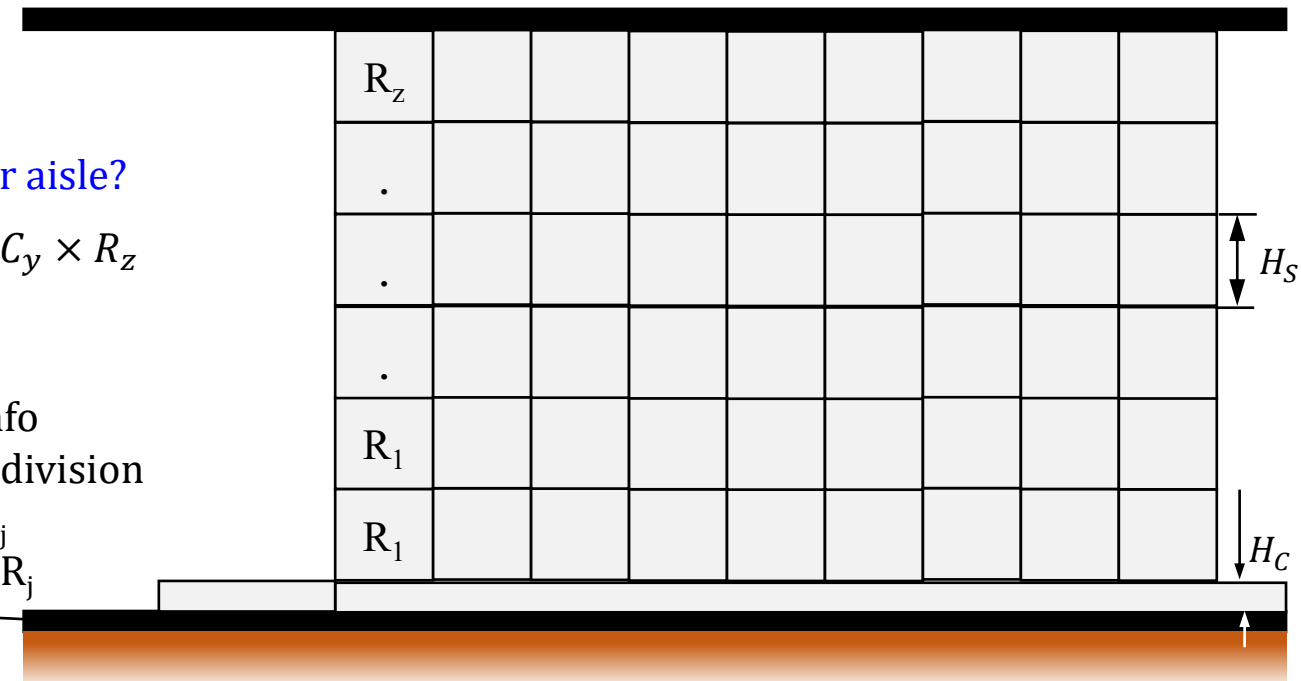
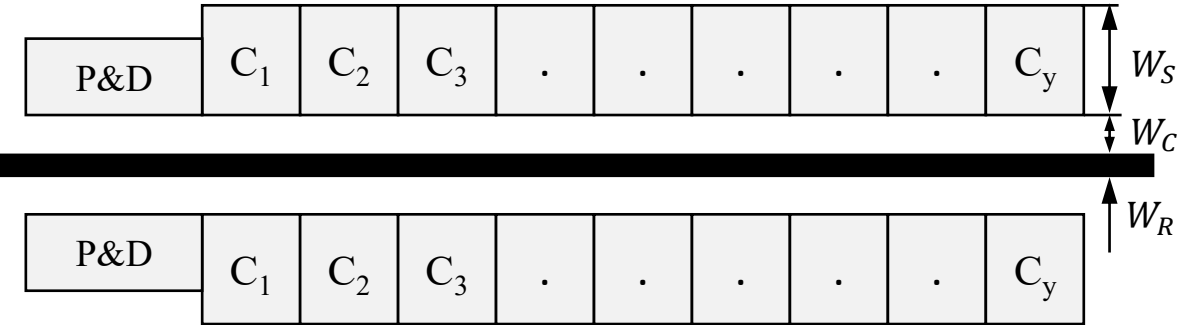
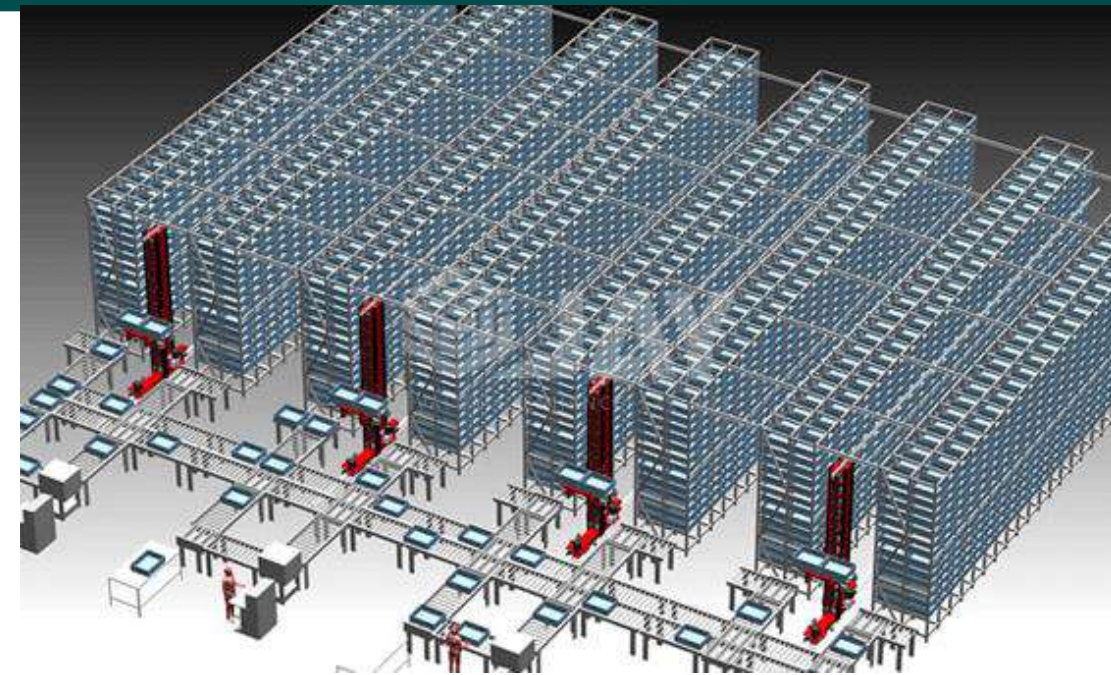


3 aisles AS/RS



Multiple aisles AS/RS

Automatic Storage and Retrieval System (AS/RS)

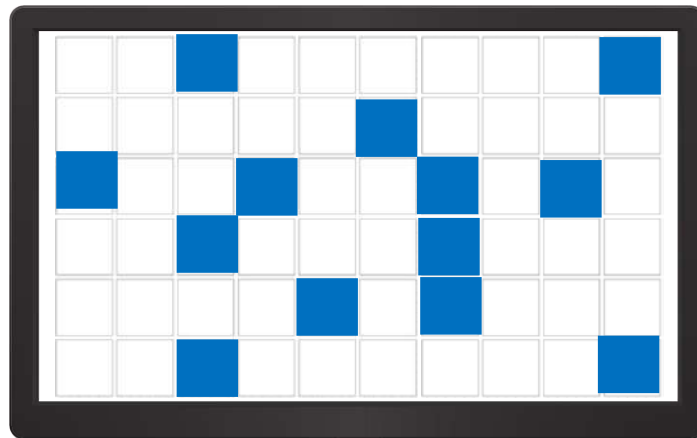


What is the capacity per aisle?

$$\text{Storage capacity} = 2 \times C_y \times R_z$$

Integrated module

- Real-time storage info
- Compartment-wise division
- Storage through $C_i R_j$
- Retrieval through $C_i R_j$



Materials handling at the workplace

- Handling of the materials after it has been delivered for use at the workplace and before it is again picked up.
- Such handlings appear to be a minor part of the overall system and does not always involve large or expensive materials handling equipment.
- However, handlings of individual jobs often take more time than for processing.
- Industrial robot can be deployed for such materials handling activities.
- Manufacturing, assembly, logistics, recycling, food, etc.

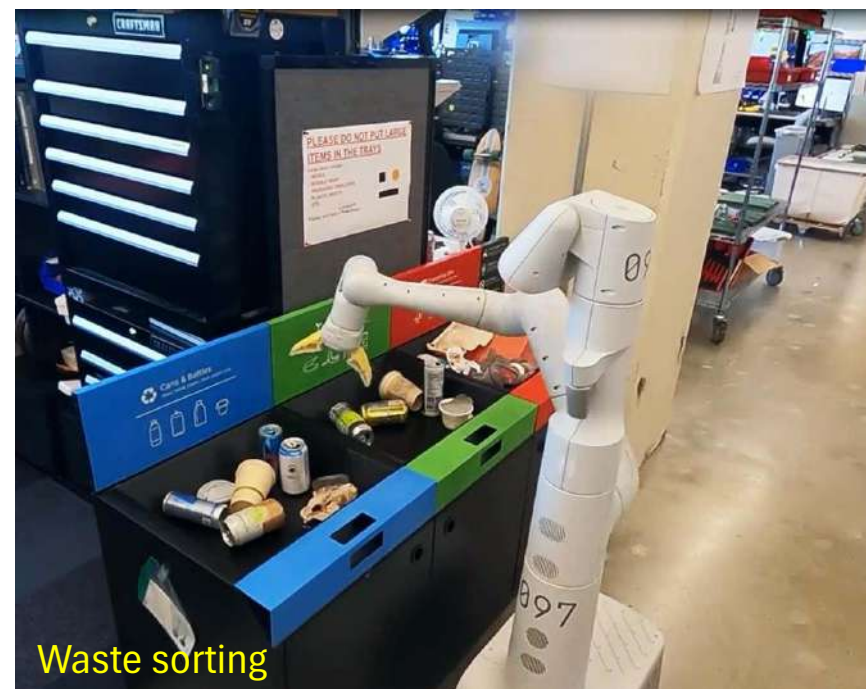


Manual handling at workplace might look trivial but such activities are monotonous, fatiguing, and also involve personal safety





Deployment of robot for MH in various industries



Two types of industrial robots: Mobile robot

Mobile Robot: Mobile robots are equipped with wheels, legs, or other locomotion mechanisms that allow them to move around in their environment. They can navigate and adapt to different spaces and are not fixed in one location.

Autonomous Vehicles



Drones



Two types of industrial robots: Stationary Robot

Stationary Robot: Performs value-adding tasks from a fixed location. These robots typically operate in a specific area and perform tasks within that designated space. Can be attached/mounted to the floor, ceiling, or other surfaces.

Floor mounted robots



Advantages of stationary robot

- **Higher Precision and Stability:** Stationary robots, like robotic arms, are firmly anchored to a fixed position, providing a high level of stability and precision for repetitive tasks.
- **Reduced Complexity:** They tend to have simpler designs compared to mobile robots because they don't need advanced navigation, sensors, or mobility systems.

Side-wall mounted robots



Disadvantages of stationary robot

- **Limited Flexibility:** Stationary robots cannot adapt to tasks outside their designated area or move between different workstations without additional equipment. This makes them less flexible compared to mobile robots.
- **Space Blockage:** They need designated space within a facility. A finite floor space gets blocked prohibiting utilization of same space for other purposes.
- **Setup Constraints:** These robots need a fixed infrastructure, such as power and communication lines, which may restrict where they can be placed in a facility and make them harder to re-deploy.



Top-mounted Stationary Robot

Top-loader/gantry robots



Advantages of top-mounted robot

- **Floor Space Utilization:** Frees up valuable floor space, making more room for other equipment, storage, or production lines for installation at floor.
- **Ergonomics and Safety:** They can operate above high-traffic areas, reducing the risk of accidents.
- **Room for other equipment:** It can operate without interfering with other floor-level systems such as conveyors, material handling systems, or human workers.

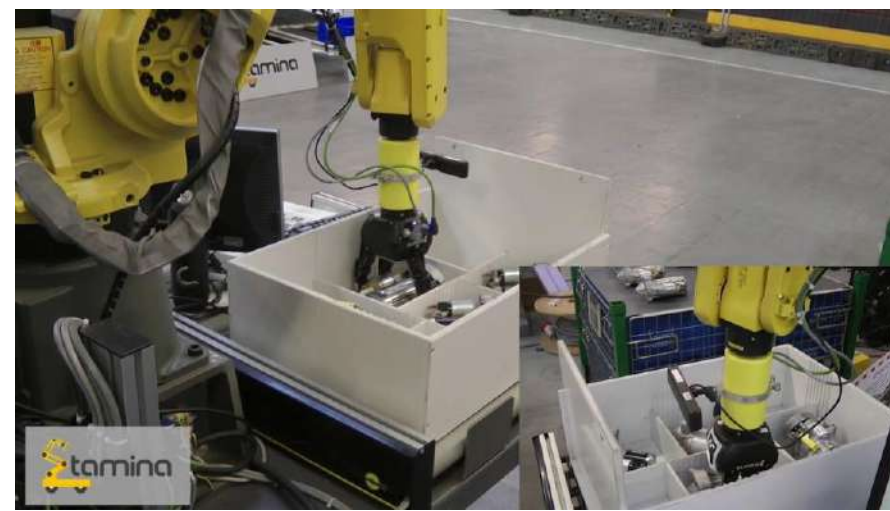
Disadvantages of top-mounted robot

- **Installation complexity:** Top-mounted robots often requires specialized structural supports, such as overhead beams, tracks, or ceiling mounting points.
- **Structural limitation:** If the ceiling or mounting points are not designed to handle the robot's weight and movement, there is a risk of structural failure.
- **Interference with other overhead systems:** Already installed lights, ducts, wires, overhead railways, or other machinery impose the risk of collision or interference.

Grouping workplace MH activities



Preparatory: Unwrapping, unpacking cleaning, sorting, re-arranging, etc.



Feeding: Placing or directing the material closer to the point-of-use



Positioning: Orienting and placing into exact location (fixture)



Manipulating: Handling the materials during actual operation/process



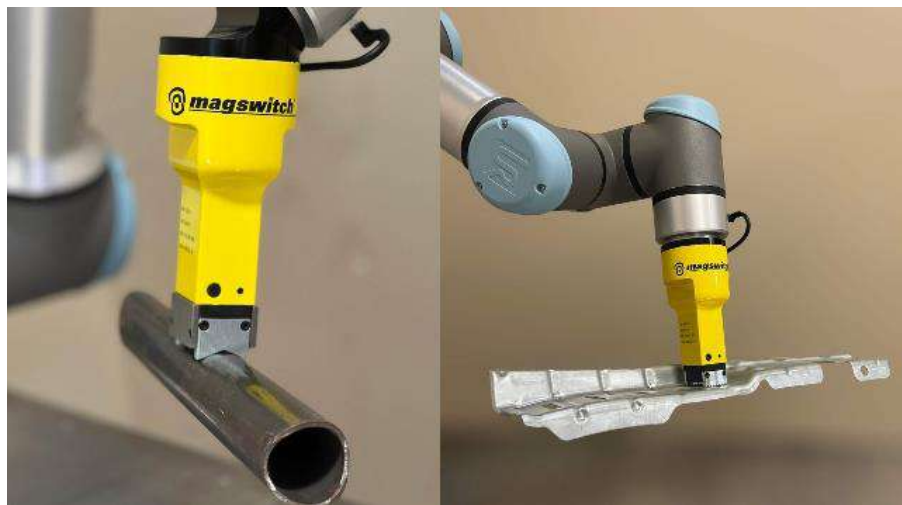
Removing: Taking out of the fixture



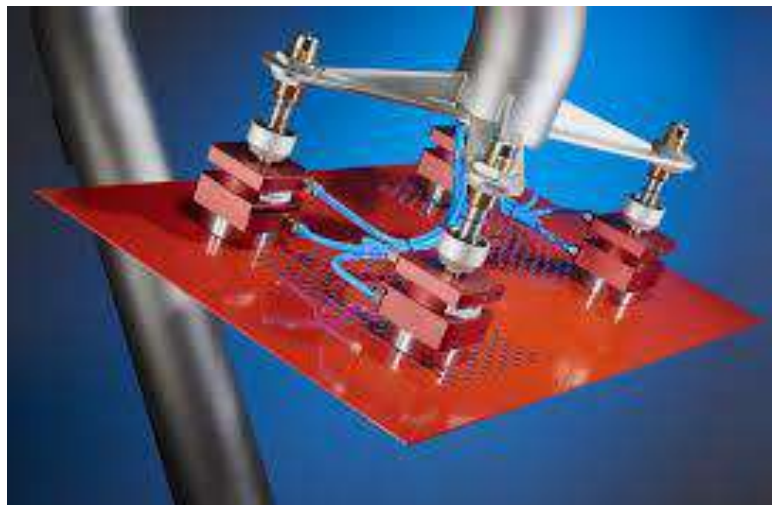
Transporting: Moving material out of the workplace

Grippers for industrial robot

A gripper is a mechanical device designed to grasp, hold, manipulate, or transport objects. It serves as the “hand” or end-effector of a robotic arm or automation system, allowing the robot to interact with the physical world by gripping and releasing objects.



Electro-magnetic gripper



Pneumatic gripper



Hydraulic gripper

Mechanical gripper



2-finger end-effector



3-finger end-effector

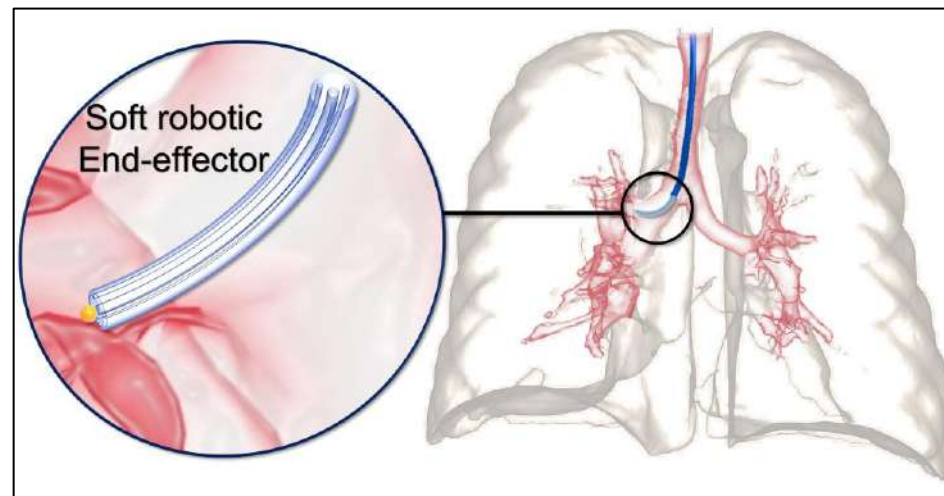
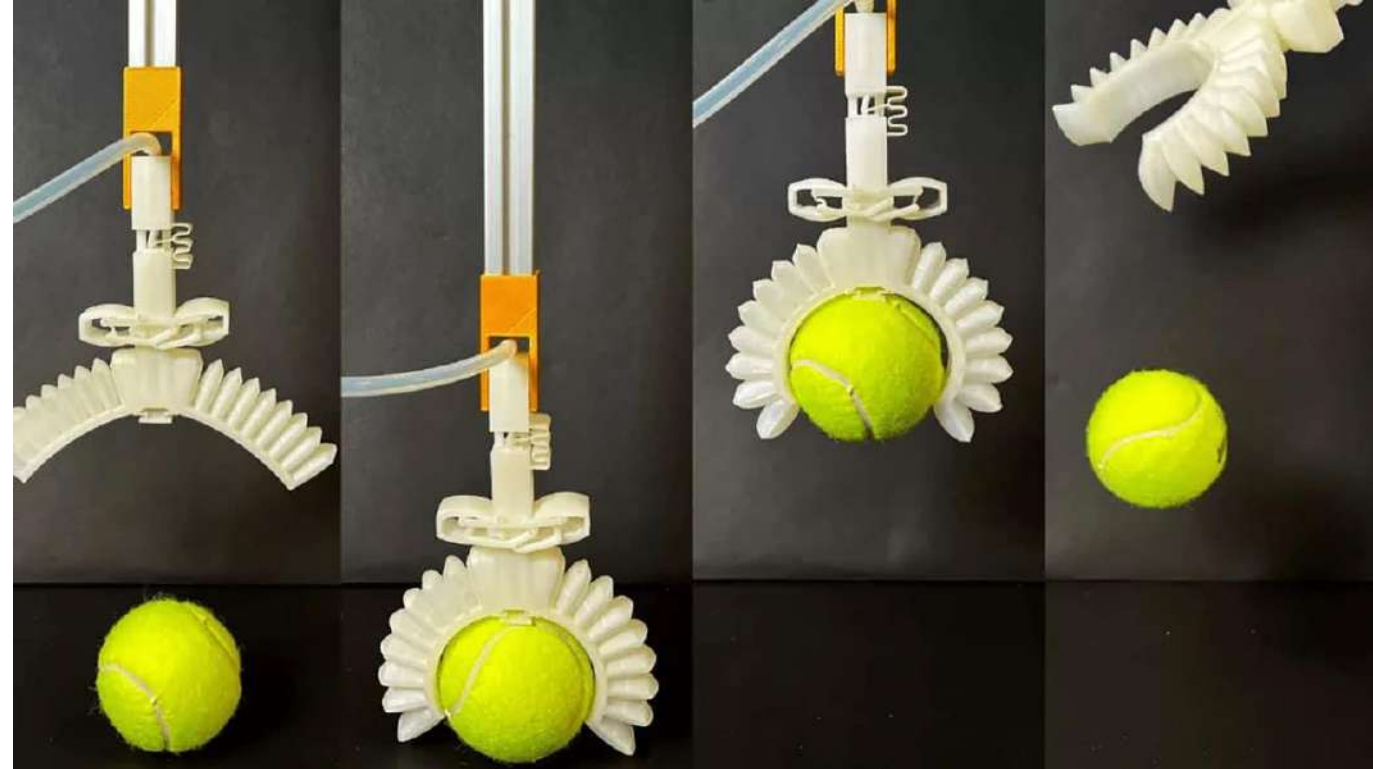


4-finger end-effector



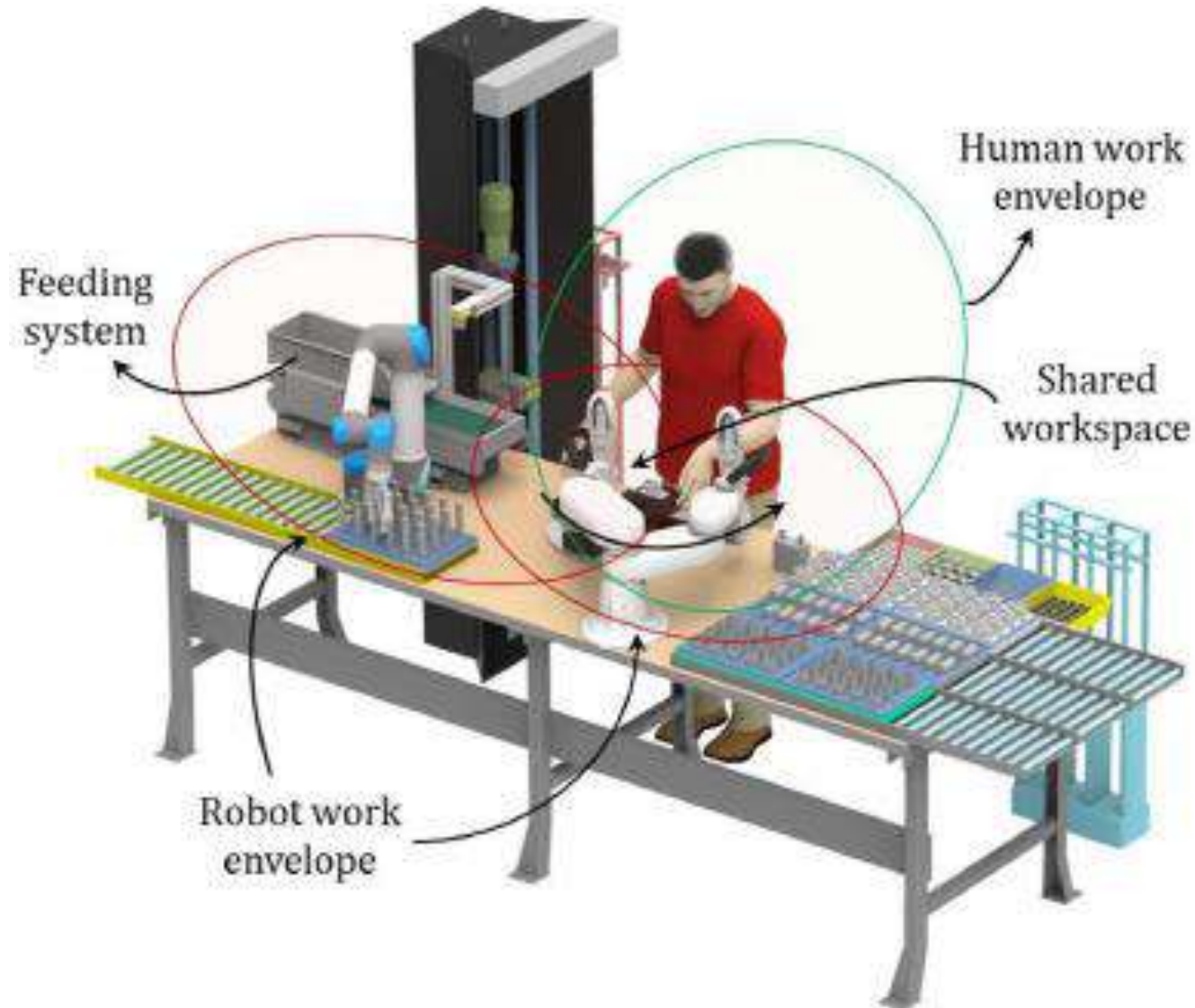
4-finger end-effector
(dexterous hand)

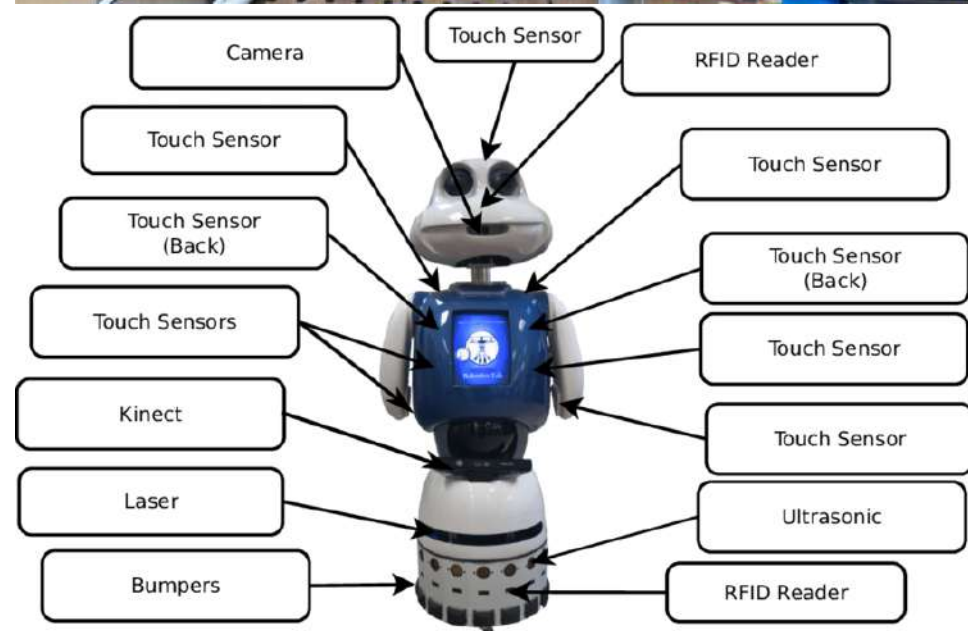
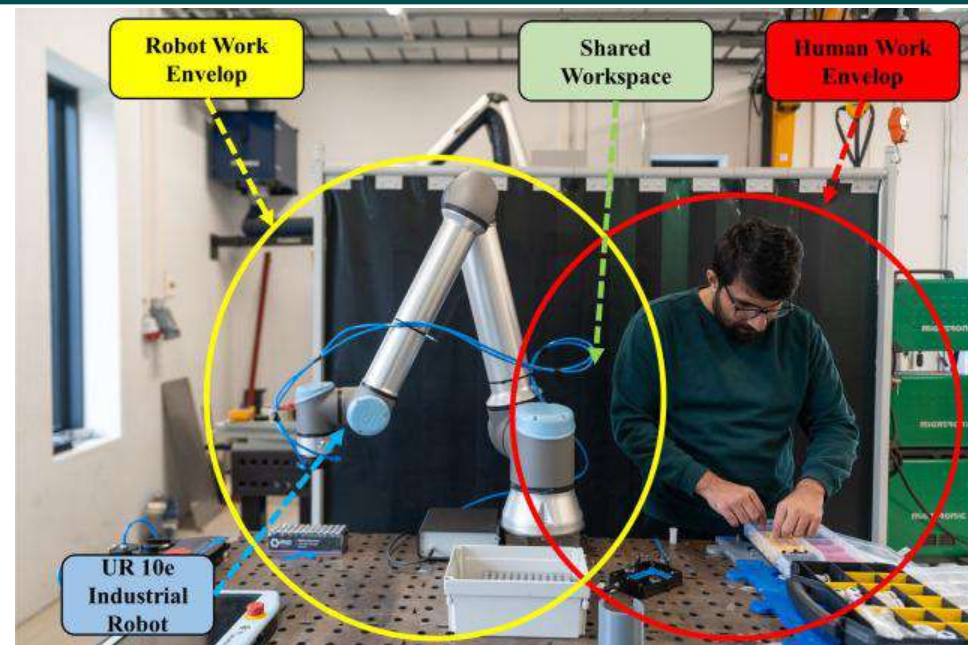
Soft grippers for robotic MH



Collaborative Robots (Cobots)

Collaborative Robots (Cobots) are a type of robot designed to work alongside humans in a shared workspace. Unlike traditional industrial robots, which are typically separated from humans due to safety concerns, cobots are built with advanced safety features that allow them to interact directly with human workers.





Key features of cobots

Safety features

- **Force Sensors:** Cobots are equipped with force sensors that can detect when they come into contact with a human or an object. If they encounter resistance, they can stop or adjust their movements, ensuring safe interaction.
- **Lightweight Design:** Cobots are typically designed to be lightweight and easy to move. This design allows for safe operation in close proximity to humans and reduces the risk of injury.
- **Speed Limiting:** Cobots can be programmed to limit their speed, ensuring that their movements remain safe and controlled when working alongside human workers.

Autonomy and collaboration

- **Autonomous Movement:** While cobots are designed to collaborate with humans, they are also capable of performing tasks autonomously. They can move materials, carry out simple actions, and respond to environmental cues without constant human intervention.
- **Shared Workspaces:** Cobots are specifically designed to share workspaces with human operators. They can assist with tasks that require both human dexterity and robotic strength, leading to a more efficient work environment.

Automated Mobile Robots (AMRs) are a class of robots designed to autonomously navigate and transport materials across a facility. AMRs use advanced sensors, cameras, and artificial intelligence to understand and adapt to their surroundings, allowing them to navigate dynamic, unstructured environments. In material handling, AMRs offer flexibility, scalability, and efficiency while reducing the need for human labor in routine tasks like transporting, delivering, and organizing materials.



Key Features of AMRs

Autonomous Navigation

- **Sensors and Mapping:** AMRs are equipped with various sensors to detect obstacles and map their environment. This allows them to move autonomously without the need for fixed tracks or paths.
- **Path Planning and AI:** AMRs use advanced algorithms and machine learning to plan their optimal routes in real-time, avoiding obstacles and dynamically adjusting their paths based on changes in the environment (e.g., moving objects, new obstacles).

Flexible Operation

- **Dynamic Adaptability:** AMRs can adapt to changes in their environment, including dynamic obstacles or layout changes in the facility.
- **Multiple Task Handling:** AMRs can be programmed to handle a variety of tasks, such as picking up materials, delivering goods, transporting products, and returning them to designated areas.

Navigation in Complex Environments

- **Obstacle Avoidance:** AMRs can autonomously detect and avoid obstacles in real-time, even in crowded or cluttered spaces, ensuring smooth material handling in dynamic environments.
- **Simultaneous Localization and Mapping (SLAM):** Using SLAM technology, AMRs can create and update a map of their environment while simultaneously keeping track of their position. This enables them to navigate accurately in unfamiliar or changing spaces.

Applications

Warehouse Automation: AMRs are widely used in logistics and warehouses for transporting goods, managing inventories, and optimizing workflows.

Delivery: AMRs are employed for last-mile deliveries, bringing packages from a distribution hub to the customer's doorstep.

Manufacturing: In factories, AMRs assist in transporting materials between production lines, improving efficiency and reducing manual labor.

Healthcare: AMRs are used in hospitals to transport medication, equipment, and supplies between departments.





Worldwide robot manufacturer



ABB Robotics (Switzerland)



KUKA Robotics (Germany)



Fanuc Corporation (Japan)



Yaskawa Electric Corporation (Japan)



Universal Robots (Denmark)





Increased Productivity: Robots can operate continuously without breaks, fatigue, or downtime, leading to significantly higher production rates compared to manual welding. They can perform repetitive tasks quickly and accurately.

Consistent Quality: Robots provide high precision and repeatability, resulting in fewer defects and higher quality welds. This consistency is particularly important in industries where structural integrity is critical, such as automotive and aerospace.

Improved Safety: Welding involves hazardous environments, including exposure to intense heat, fumes, and UV radiation. By automating welding processes, companies can reduce workers' exposure to these dangers, thereby improving overall workplace safety.

Cost Reduction: While the initial investment in robotic welding systems can be considerable, the long-term savings from reduced labor costs, fewer errors, and decreased material waste can be substantial. Robots can also help in optimizing the use of welding materials.

Flexibility and Adaptability: Modern robotic systems can be programmed and reconfigured for different welding tasks and materials. This flexibility allows manufacturers to adapt quickly to changes in production requirements or to switch between products without extensive downtime.

Enhanced Capabilities: Robots can perform complex welding techniques that may be difficult to achieve manually. Advanced welding processes, such as laser welding or automated TIG and MIG welding, can be executed effectively with robotic systems.

Remote Monitoring and Data Collection: Robotic welding systems can be integrated with sensors and IoT technology, allowing for real-time monitoring and data collection. This capability helps in predictive maintenance, quality control, and process optimization.

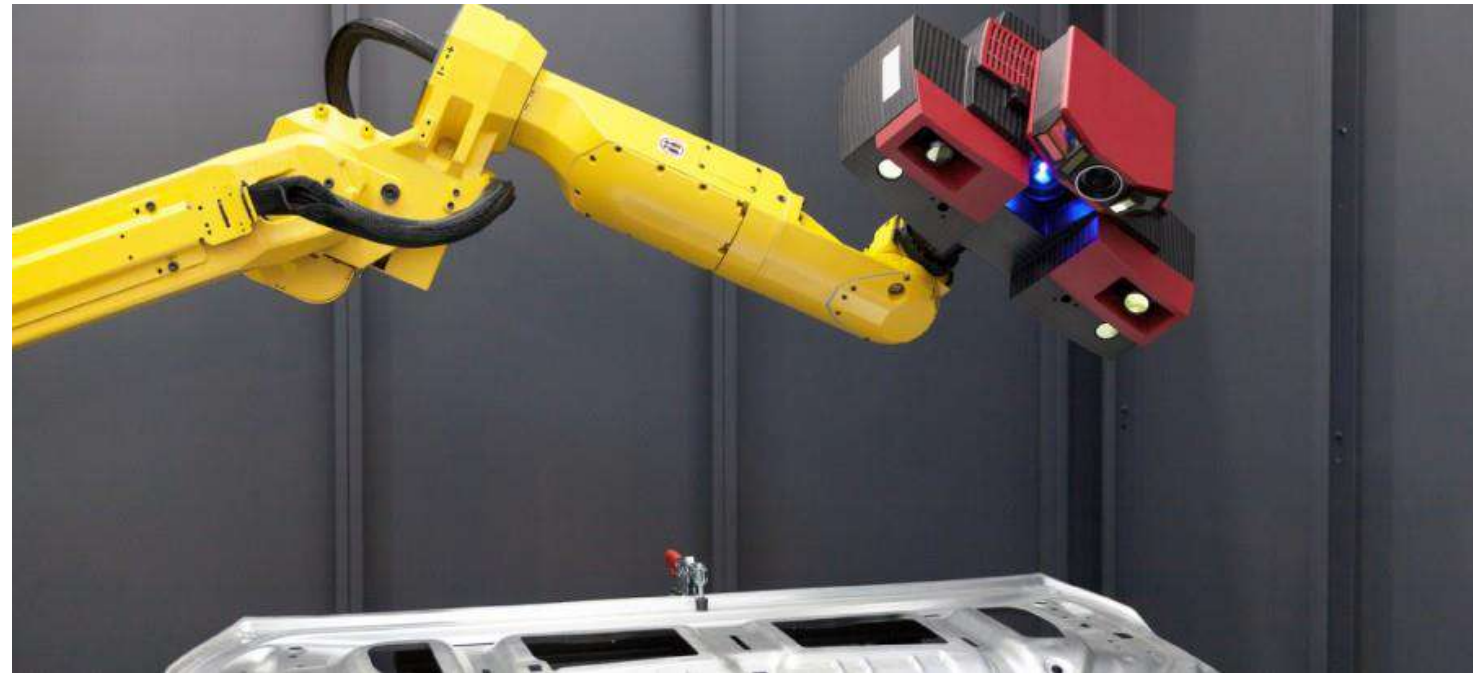
Workforce Shortage Mitigation: The manufacturing industry often faces challenges related to labor shortages and finding skilled workers. Robots can help fill this gap by performing tasks that may be difficult to staff with qualified personnel.

Integration with Other Automation Systems: Robotic welding can be integrated with other automation processes, such as material handling and inspection systems, leading to streamlined operations and a more cohesive production line.

Reduced Welding Distortion: Robots can apply consistent heat and control the welding parameters precisely, which can help mitigate the warping and distortion of materials often associated with manual welding.

Benefits of Robotic Inspection

- **Increased Efficiency:** Robots can perform inspections faster than human operators, reducing downtime and improving productivity.
- **High Accuracy & Consistency:** Robotic systems provide precise and repeatable measurements, ensuring that defects are consistently detected and evaluated.
- **Enhanced Safety:** Dangerous or difficult-to-reach inspection areas, such as high-temperature environments or confined spaces, can be inspected by robots, eliminating safety risks to human workers.
- **Cost Savings:** Over time, robotic inspection reduces labor costs and minimizes human error, which can result in costly repairs or re-work.
- **Automation & Integration:** Robotic systems can be integrated into automated production lines, enabling continuous and real-time weld inspection without interrupting production.



Robotic Inspection Technologies

- **Vision-Based Inspection:** Robots use cameras and image-processing algorithms to detect surface defects such as cracks, weld spatter, etc.
- **Ultrasonic Testing (UT):** Robots equipped with ultrasonic probes can detect subsurface defects like voids, cracks, or porosity.
- **Eddy Current Testing (ECT):** ECT systems are used for detecting surface and near-surface defects, particularly in conductive materials.
- **Laser Scanning:** Robots with laser sensors can create detailed 3D models of the weld and surrounding areas, identifying discrepancies in weld geometry.

Thank you